

```
!pip install mlxtend
```

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Requirement already satisfied: mlxtend in c:\users\dell\anaconda3\lib\site-  
packages (0.23.3)  
Requirement already satisfied: scipy>=1.2.1 in  
c:\users\dell\anaconda3\lib\site-packages (from mlxtend) (1.13.1)  
Requirement already satisfied: numpy>=1.16.2 in  
c:\users\dell\anaconda3\lib\site-packages (from mlxtend) (1.26.4)  
Requirement already satisfied: pandas>=0.24.2 in  
c:\users\dell\anaconda3\lib\site-packages (from mlxtend) (2.2.2)  
Requirement already satisfied: scikit-learn>=1.3.1 in  
c:\users\dell\anaconda3\lib\site-packages (from mlxtend) (1.5.1)  
Requirement already satisfied: matplotlib>=3.0.0 in  
c:\users\dell\anaconda3\lib\site-packages (from mlxtend) (3.9.2)  
Requirement already satisfied: joblib>=0.13.2 in  
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Requirement already satisfied: contourpy>=1.0.1 in  
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(1.2.0)  
Requirement already satisfied: cycler>=0.10 in  
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(0.11.0)  
Requirement already satisfied: fonttools>=4.22.0 in  
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(4.51.0)  
Requirement already satisfied: kiwisolver>=1.3.1 in  
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(1.4.4)  
Requirement already satisfied: packaging>=20.0 in  
c:\users\dell\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend)  
(24.1)  
Requirement already satisfied: pillow>=8 in c:\users\dell\anaconda3\lib\site-  
packages (from matplotlib>=3.0.0->mlxtend) (10.4.0)  
Requirement already satisfied: pyparsing>=2.3.1 in  
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(3.1.2)  
Requirement already satisfied: python-dateutil>=2.7 in  
c:\users\dell\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend)  
(2.9.0.post0)  
Requirement already satisfied: pytz>=2020.1 in  
c:\users\dell\anaconda3\lib\site-packages (from pandas>=0.24.2->mlxtend)  
(2024.1)  
Requirement already satisfied: tzdata>=2022.7 in  
c:\users\dell\anaconda3\lib\site-packages (from pandas>=0.24.2->mlxtend)  
(2023.3)  
Requirement already satisfied: threadpoolctl>=3.1.0 in  
c:\users\dell\anaconda3\lib\site-packages (from scikit-learn>=1.3.1->mlxtend)  
(3.5.0)  
Requirement already satisfied: six>=1.5 in c:\users\dell\anaconda3\lib\site-  
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->mlxtend) (1.16.0)
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# Import Libraries
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

# =====
# Load Dataset
# =====
file_path = r"C:\Users\Dell\Documents\Modified_Crime_Dataset.csv"
data = pd.read_csv(file_path)

# =====
# Preprocessing: Binary Encoding
# =====
# Select relevant columns (e.g., crime categories)
crime_columns = data.columns[3:] # Adjust column range based on your dataset
                                   structure

# Binarize data: Convert non-zero values to 1 (True) and zero to 0 (False)
binary_data = (data[crime_columns] > 0).astype(bool)

# Debug: Print sample of binary data
print("Binary Data Sample:")
print(binary_data.head())

# =====
# Dimensionality Reduction
# =====
# Reduce columns to top features with high variance (optional)
from sklearn.feature_selection import VarianceThreshold

selector = VarianceThreshold(threshold=0.01) # Adjust threshold if needed
binary_data_reduced = selector.fit_transform(binary_data)

# Debug: Print reduced dataset shape
print("Reduced Binary Data Shape:", binary_data_reduced.shape)

# =====
# Apply Apriori Algorithm
# =====
# Set a higher min_support to reduce memory usage
frequent_itemsets = apriori(pd.DataFrame(binary_data_reduced),
                             min_support=0.1, # Adjust as needed (higher
                                                reduces memory usage)
                             use_colnames=True,
                             low_memory=True)

# Debug: Print frequent itemsets
print("Frequent Itemsets:")
print(frequent_itemsets)

# =====
# Generate Association Rules

```

```
# =====
rules = association_rules(frequent_itemsets, metric="confidence",
                          min_threshold=0.7)

# Debug: Print association rules
print("Association Rules:")
print(rules)

# =====
# Save Results
# =====
# Save frequent itemsets and rules to CSV files
frequent_itemsets.to_csv(r"C:\Users\Dell\Documents\frequent_itemsets.csv",
                          index=False)
rules.to_csv(r"C:\Users\Dell\Documents\association_rules.csv", index=False)

print("Frequent itemsets and rules saved to CSV files.")
```

Binary Data Sample:

	CULPABLE HOMICIDE NOT AMOUNTING TO MURDER	RAPE	CUSTODIAL RAPE	\
0	True	True	False	
1	True	True	False	
2	True	True	False	
3	True	True	False	
4	True	True	False	

	OTHER RAPE	KIDNAPPING & ABDUCTION	\
0	True	True	
1	True	True	
2	True	True	
3	True	True	
4	True	True	

	KIDNAPPING AND ABDUCTION OF WOMEN AND GIRLS	\
0	True	
1	True	
2	True	
3	True	
4	True	

	KIDNAPPING AND ABDUCTION OF OTHERS	DACOITY	\
0	True	True	
1	True	True	
2	True	True	
3	True	True	
4	True	True	

	PREPARATION AND ASSEMBLY FOR DACOITY	ROBBERY	...	DISTRICT_WASHIM	\
0	False	True	...	False	
1	False	True	...	False	
2	False	True	...	False	
3	False	True	...	False	
4	False	True	...	False	

	DISTRICT_WAYANADU	DISTRICT_WEST	DISTRICT_WEST	GODAVARI	DISTRICT_WOKHA
\					
0	False	False		False	False
1	False	False		False	False
2	False	False		False	False
3	False	False		False	False
4	False	False		False	False

	DISTRICT_YADGIRI	DISTRICT_YAMUNANAGAR	DISTRICT_YAVATMAL	\
0	False	False	False	
1	False	False	False	
2	False	False	False	
3	False	False	False	
4	False	False	False	

	DISTRICT_ZUNHEBOTO	TOTAL_RAPE
0	False	True
1	False	True
2	False	True
3	False	True
4	False	True

[5 rows x 868 columns]

Reduced Binary Data Shape: (9017, 51)